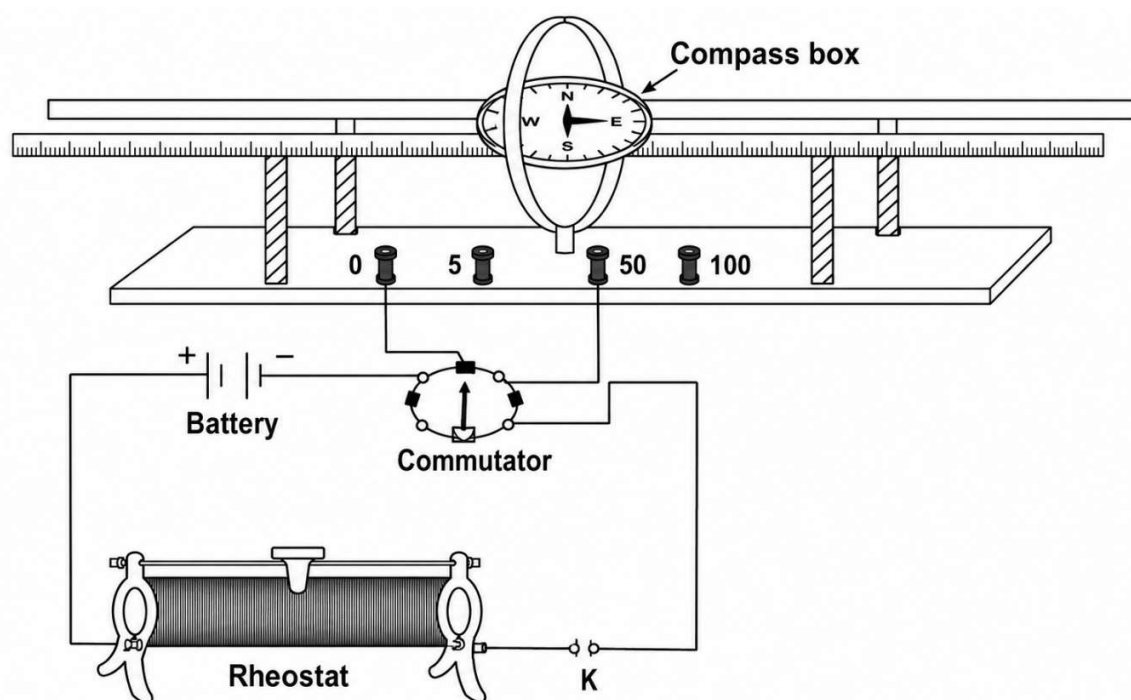


IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: To study the variation of magnetic field with the distance along the axis of a current carrying coil.		
EXPERIMENT NO. 06	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 4

AIM: To study the variation of magnetic field with the distance along the axis of a current carrying coil.

APPARATUS: Stewart and Gee type tangent galvanometer, a battery or power supply, rheostat, Commutator, four ways key, compass box, plugs key and connecting wires.

CIRCUIT DIAGRAM:



THEORY AND FORMULA USED:

If a current-carrying coil is placed in the y - z plane, its axis will align with the x -axis. The magnetic field along the coil's axis is expressed as:

$$B = \frac{\mu_0 n i x r^2}{2H(r^2 + x^2)^{3/2}}$$

Where, μ_0 ($= 4\pi \times 10^{-7}$ H/m) is the vacuum permeability, N is the number of turns of the field coil, I is the current in the wire, r is the radius of the coil and x is the axial distance from the center of the coil. If θ is the deflection produced in magnetometer at a certain position on the axis of coil, then magnetic field at that point will be,

$$B = H \tan \theta$$

The above equation implies that the graph between x and $\tan \theta$ will give the variation of magnetic field at the axis of circular coil.

PROCEDURE:

1. Place the instrument in such a way that the arms of the magnetometer lie roughly east and west and the magnetic needle lies at the center of the vertical coil. Place the eye a little above the coil and rotate the instrument in the horizontal plane till the coil, the needle and its image in the mirror provided at the base of the compass box, all lie in same vertical plane. The coil is thus set roughly in the magnetic meridian. Rotate the compass box so that the pointer lies on the 0-0 line.
2. Connect all the components as shown in circuit diagram.
3. Adjust the value of the current so that the magnetometer at central position gives a deflection of the order of 70° - 75° . Note this magnetometer reading for the both directions of currents. This will give you θ value at $x=0$.
4. Now slide the magnetometer along the $+x$ axis of coil with an increment of 3 cm and note the deflection of needle in magnetometer (both ends of needle position) for the both directions of current in coil. Record a number of observations. ($x = 0, 2, 4, 6, 8, 10, 12$ cm)
5. After this, repeat the point 4 for the magnetometer position along $-x$ axis of coil. *i.e.* repeat the observation by shifting the magnetometer in the opposite direction and keeping the current constant at the same value.

OBSERVATIONS:

- a. Least count of the magnetometer =

- b. Current $I =$
 c. Number of turn sin coil $n =$
 d. Radius of the coil $=$

OBSERVATION TABLE:

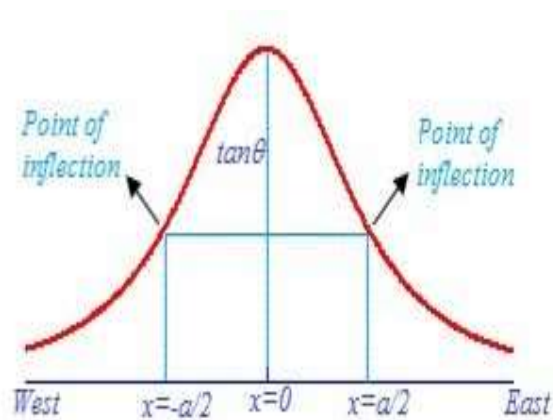
Table A: Deflection in magnetometer along +axis of coil.

Sr. No.	Distance of needle from center, x (cm)	Deflection on East Arm				tanθ	
		Current in one direction		Current in reverse direction			Mean θ (in deg.)
		θ ₁	θ ₂	θ ₃	θ ₄		

Table B: Deflection in magnetometer along -axis of coil.

Sr. No.	Distance of needle from center, x (cm)	Deflection on East Arm				tanθ	
		Current in one direction		Current in reverse direction			Mean θ (in deg.)
		θ ₁	θ ₂	θ ₃	θ ₄		

The plot of $\tan\theta$ vs x will be appears as follow:



RESULT:

With help of the graph between $\tan \theta$ and x , following points can be concluded.

1. The intensity of magnetic field is maximum at the center and goes on decreasing as we move away from the center of the coil towards right or left.
2. The point on the both side of graph where curve becomes convex to concave (i.e. the curve changes its nature) are called the point of inflection. The distance between the two points of inflection is equal to the radius of the circular coil.
3. The radius of coil= distance between points of inflection =cm. The magnetic field intensity is maximum at the center of the coil and decreasing away from the center of the coil either towards right or left.

PRECAUTIONS:

1. There should be no magnet, magnetic substances and current carrying conductor near the apparatus.
2. The plane of the coil should be set in the magnetic medium.
3. The current should remain constant and should be reversed for each observation. The readings should be taken by viewing vertically on apparatus.

Viva Questions:

1. What is the magnetic field along the axis of a circular current-carrying coil?
2. Why does the magnetic field decrease symmetrically on either side of the center?

3. How does the current in the coil affect the magnetic field?
4. What would happen to the field distribution if the coil radius is increased?
5. How does this experiment demonstrate the principle of superposition of magnetic fields?